How-To Library

Informatica

Importing Users From Different Kerberos Realms in Test Data Management

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Abstract

Users from multiple Kerberos realms can access and use Test Data Management (TDM) that you install in a Kerberos environment. This article provides the steps to add users from one Kerberos realm to TDM installed in another realm and assign Test Data Manager Service privileges.

Supported Versions

- Test Data Management 10.2.0 HotFix 2

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Overview

You can install TDM in a Kerberos authentication environment. You can also use TDM in an Informatica domain configured to use Kerberos cross-realm authentication.

Kerberos cross-realm authentication enables Informatica clients that belong to one Kerberos realm to authenticate with nodes and application services that belong to another Kerberos realm. So, a TDM user created in one Kerberos realm can access TDM that you install in another realm.

To allow users from one Kerberos realm to access a Test Data Manager Service created in another realm, set up a connection to the LDAP server. Specify the users and groups that can have access to the Informatica domain. Then import the user accounts into LDAP security domains. You can then assign the privileges required to perform TDM tasks.

Example

You work for a test team that uses Test Data Management to create and manage test data.

For a specific project you need to allow users from a different QA team to use your Test Data Manager Service to run plans. You install TDM in a Kerberos authentication environment, and the organization includes multiple Kerberos realms. Different QA teams belong to different realms with cross-realm communication enabled. You create the Test Data Manager Service in an Informatica domain in your security domain. All users in your realm belong to this security domain. You need to allow users from the *PLATFORMKRB.COM* Kerberos realm to access your Test Data Manager Service and run plans.

To import users and user groups from a different Kerberos realm and add them as TDM users and user groups, perform the following tasks:

1. Configure the domain and add users and groups in Informatica Administrator in your Kerberos realm.
2. Assign privileges required to log in and use Test Data Manager.
3. Verify that the user can perform required tasks.

Prerequisites

Before you import users from a different Kerberos realm, perform the following tasks:

- Verify that the Test Data Management version supports cross-realm authentication.
- Ensure that the following files are configured in the Kerberos realm where you create the Test Data Manager Service:
  - `krb5.conf`. Contains configuration information for Kerberos. This file is stored in the following location: `<INFA_HOME>/TDM/datadirect`.
  - `org.eclipse.virgo.kernel.authentication.config`. Contains authentication properties for the TDM server machine. This file is stored in the following location: `<INFA_HOME>/TDM/configuration`.

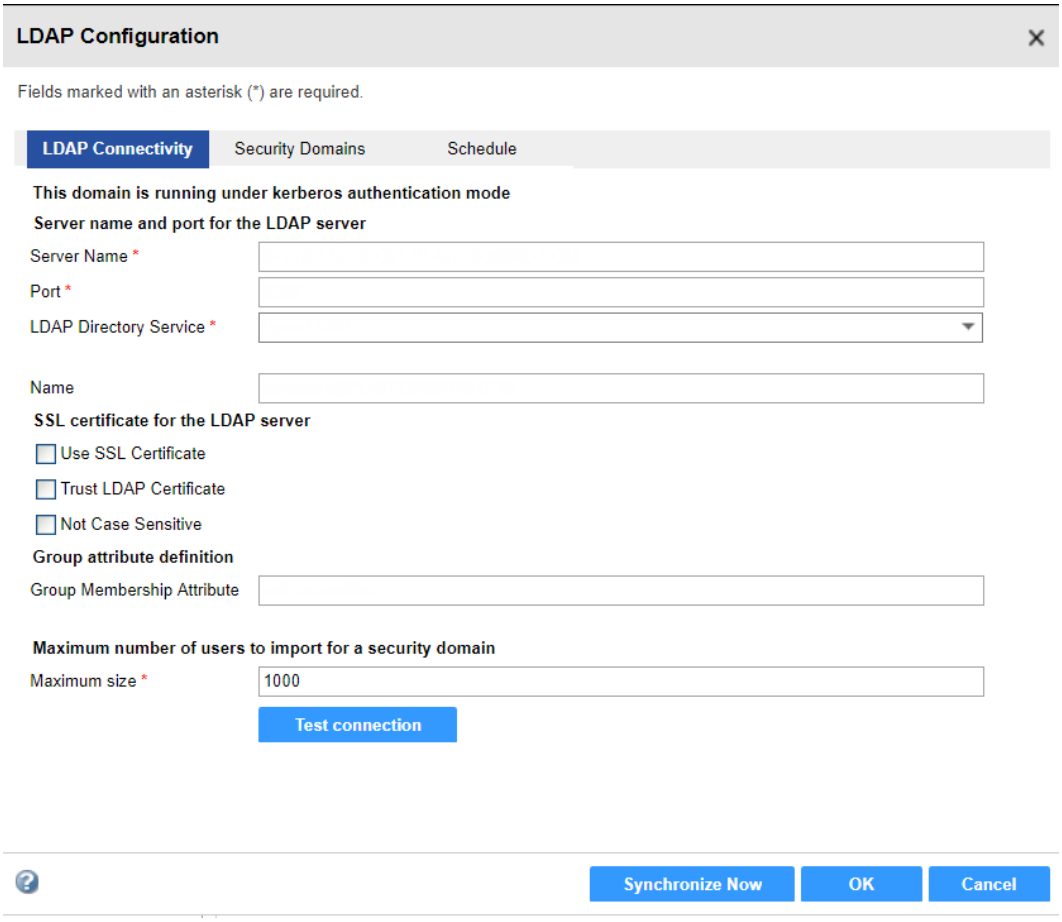
Step 1. Configure the Domain and Add Users and Groups

Configure the Kerberos domain that you want to add to the Informatica domain in which you create the Test Data Manager Service. Add users and groups from the domain to provide access.

1. Log in to the Administrator tool.

   In a Kerberos authentication environment, you choose the security domain and click **Log In**.
2. Click the **Security** tab.
3. Click the **Actions** menu and select **LDAP Configuration**.
4. In the **LDAP Configuration** dialog box, click the **LDAP Connectivity** tab.
5. Configure the connection properties for the LDAP server.

The following image shows the fields on the LDAP Connectivity tab:



You might need to consult the LDAP administrator to get the information needed to connect to the LDAP server.

The following table describes the LDAP server configuration properties:

Property	Description
Server name	Host name or IP address of the machine hosting the LDAP directory service.
Port	Listening port for the LDAP server. This is the port number to communicate with the LDAP directory service. Typically, the LDAP server port number is 389. If the LDAP server uses SSL, the LDAP server port number is 636. The maximum port number is 65535.
LDAP Directory Service	Type of LDAP directory service. **Note:** If you use Kerberos authentication, you must select Microsoft Active Directory Service.

Property	Description
Name	Distinguished name (DN) for the principal user. The user name often consists of a common name (CN), an organization (O), and a country (C). The principal user name is an administrative user with access to the directory. Specify a user that has permission to read other user entries in the LDAP directory service. To connect to Azure Active Directory, specify the User Principal Name (UPN) for the principal user.
Password	Password for the principal user. Leave blank for anonymous log in.
Use SSL Certificate	Indicates that the LDAP server uses the Secure Socket Layer (SSL) protocol.
Trust LDAP Certificate	Determines whether the Service Manager can trust the SSL certificate of the LDAP server. If selected, the Service Manager connects to the LDAP server without verifying the SSL certificate. If not selected, the Service Manager verifies that the SSL certificate is signed by a certificate authority before connecting to the LDAP server.
Not Case Sensitive	Indicates that the Service Manager must ignore case sensitivity for distinguished name attributes when assigning users to groups.
Group Membership Attribute	Name of the attribute that contains group membership information for a user. This is the attribute in the LDAP group object that contains the DNs of the users or groups who are members of a group. For example, *member* or *memberof*.
Maximum Size	Maximum number of user accounts to import into a security domain. For example, if the value is set to 100, you can import a maximum of 100 user accounts into the security domain. If the number of user to be imported exceeds the value for this property, the Service Manager generates an error message and does not import any user. Set this property to a higher value if you have many users to import. Default is 1000.

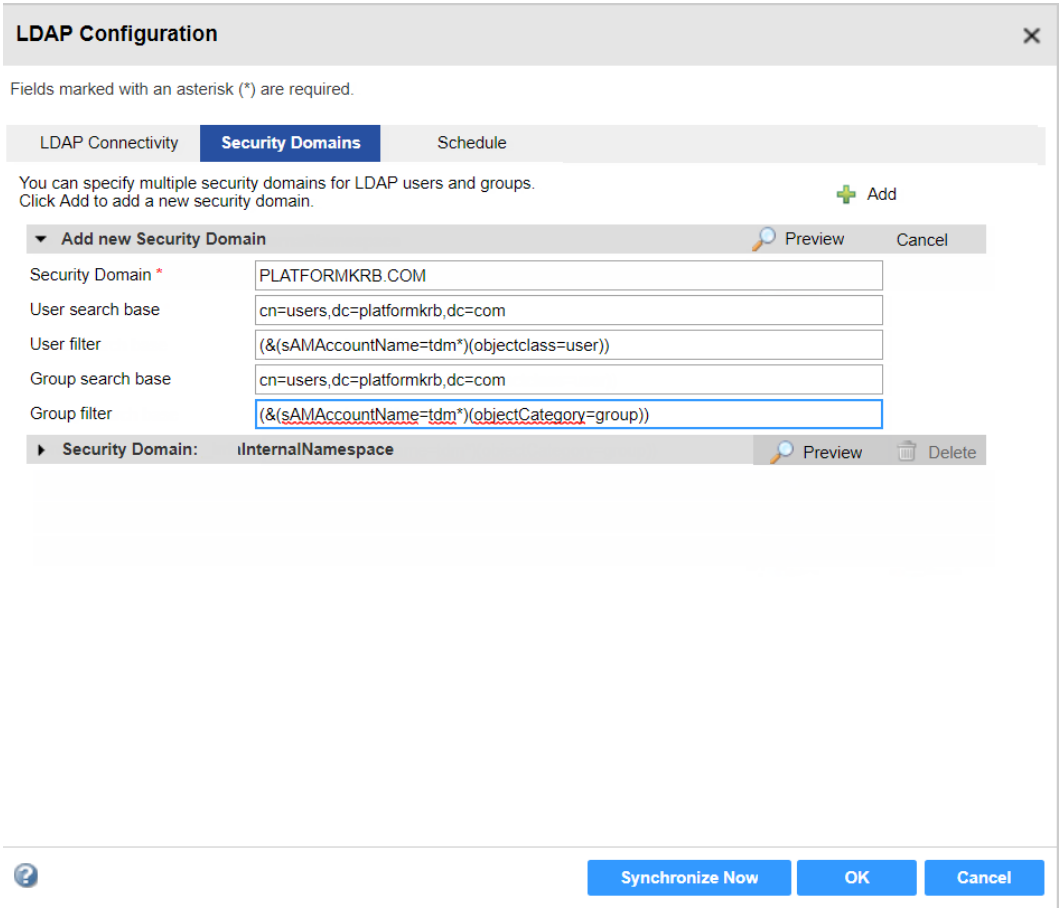
6. Click **Test Connection** to verify that the connection to the LDAP server is valid.
7. Click the **Security Domains** tab.
8. Click **Add**.

   The following table describes the properties and filters that you can set for a security domain:

Property	Description
Security Domain	Name of the LDAP security domain. The name is not case sensitive and must be unique within the domain. The string cannot exceed 128 characters or contain the following special characters: , + / < > @ ; \ % ? The name can contain an ASCII space character except for the first and last character. All other space characters are not allowed.
User search base	Distinguished name (DN) of the entry that serves as the starting point to search for user names in the LDAP directory service. The search finds an object in the directory according to the path in the distinguished name of the object. For example, in Microsoft Active Directory, the distinguished name of a user object might be cn=UserName,ou=OrganizationalUnit,dc=DomainName, where the series of relative distinguished names denoted by dc=DomainName identifies the DNS domain of the object.

Property	Description
User filter	An LDAP query string that specifies the criteria for searching for users in the directory service. The filter can specify attribute types, assertion values, and matching criteria. For example: `(objectclass=*)` searches all objects. `(&(objectClass=user)(!(cn=susan)))` searches all user objects except “susan”. For more information about search filters, see the documentation for the LDAP directory service.
Group search base	Distinguished name (DN) of the entry that serves as the starting point to search for group names in the LDAP directory service.
Group filter	An LDAP query string that specifies the criteria for searching for groups in the directory service.

The following image shows the properties for a PLATFORMKRB.COM LDAP security domain entered on the Security Domains tab of the **LDAP Configuration** dialog box:



The user filter is set to import all users beginning with "tdm". The user group filter is set to import all groups beginning with "tdm".

9. Click **Preview** to view a subset of the list of users and groups that fall within the filter parameters.

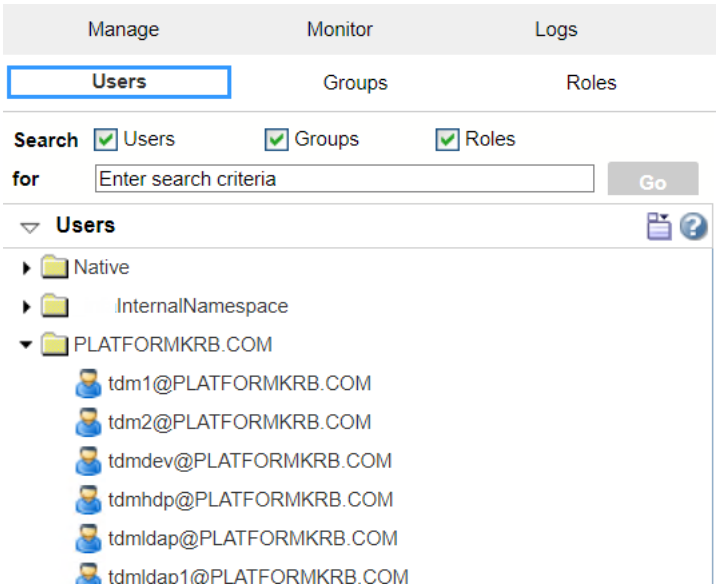
   If the preview does not display the correct set of users and groups, modify the user and group filters and search bases to get the correct users and groups.

10. To immediately synchronize the users and groups in the security domains with the users and groups in the LDAP directory service, click **Synchronize Now**.

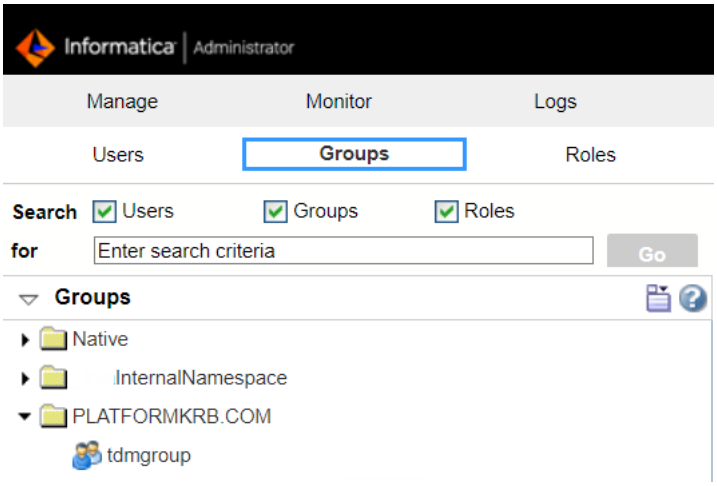
    The Service Manager synchronizes the users in all the LDAP security domains with the users in the LDAP directory service. The time it takes for the synchronization process to complete depends on the number of users and groups to be imported.

11. Click **OK** to save the security domain and add the users and groups.

    The names appear in the panel on the left under the security domain that you added.
    The following image shows the users from the security domain listed on the **Users** tab:



    The following image shows the group from the security domain listed on the **Groups** tab:



Repeat the steps for each security domain that you want to add.

Step 2. Assign Service Roles and Privileges

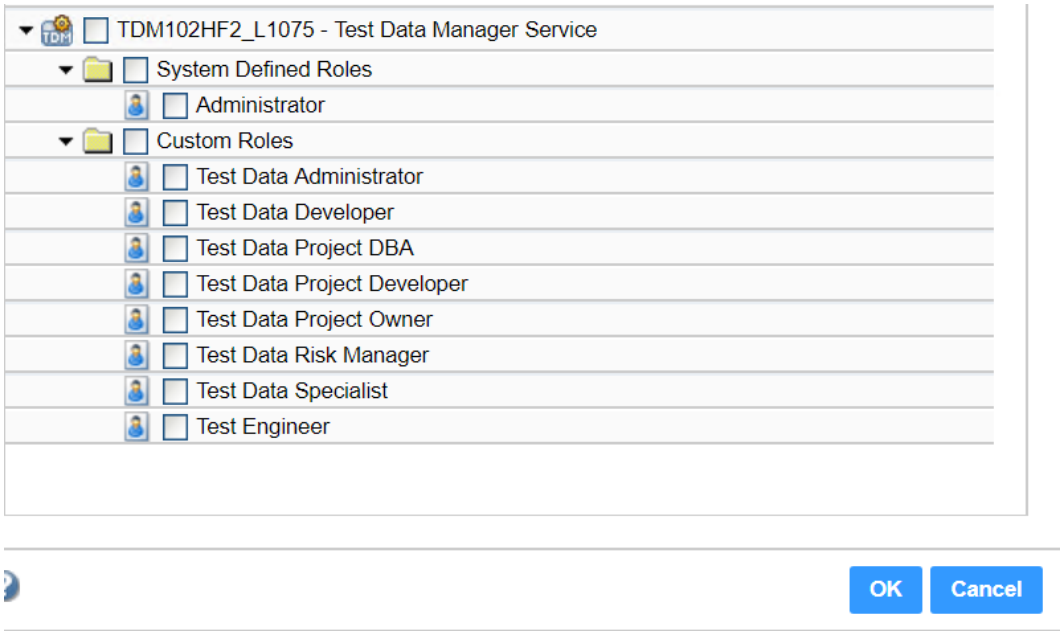
You assign privileges from the **Security** tab in Informatica Administrator. You can choose to assign a custom role to a user. When you assign a role, the user gets all default privileges associated with the role.

1. In the Administrator tool, click the **Security** tab and click **Users**.

2. In the navigator, select the PLATFORMKRB.COM security domain and expand the list of users.
3. Select a user and click **Privileges**.
4. Click **Edit**.

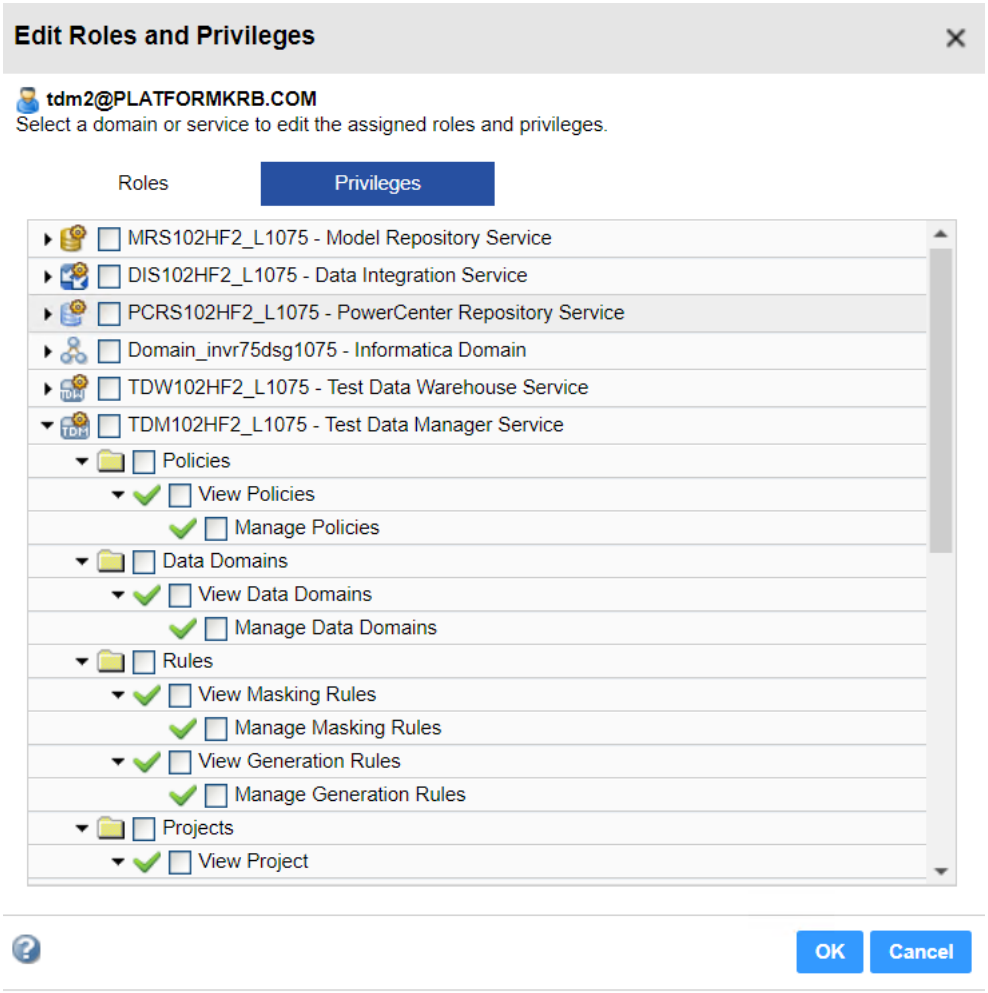
   The **Edit Roles and Privileges** dialog box appears.
5. To assign roles, expand the Test Data Manager Service on the **Roles** tab.
6. Select the Test Data Manager Service role to assign to the user.

   The following image shows the custom roles available to choose from:



7. To assign privileges, click the **Privileges** tab.
8. Expand the Test Data Manager Service.
9. Select the privileges that you want to assign to the user.

The following image shows the **Edit Roles and Privileges** dialog box:



You cannot revoke privileges inherited from a role or group.

10. Click **OK**.
11. To assign roles and privileges to a user group, on the **Security** tab click **Groups**.
12. Select the PLATFORMKRB.COM security domain and expand the list of groups.
13. Select a group and click **Privileges**.
14. Repeat steps 4 through 10 to assign roles and privileges to the user group.

Step 3. Verify User Privileges

You can verify that users from the security domain that you added can access your Test Data Manager Service and perform required tasks. Log in as the user and perform a task for which you assigned required privileges.

1. Log in to Test Data Manager.

   The security domain that you added must be visible in the list. Select the security domain and click **Log In**.
2. Open the required project.
3. Click the **Execute** tab and select the plan that you want to run.
4. Click **Actions** > **Generate and Execute**.

The **Generate and Execute Plan** window opens.

5. Choose the Integration Service.
6. Choose to run the workflow immediately.
7. Click **Generate and Execute**.

You can monitor the status of the job from the **Monitor** tab and view the data in the target after the job completes. If you encounter access issues or cannot run the job as the user, verify that you have assigned all the required privileges.

User Privileges

The users that you add to Informatica require relevant privileges to be able to perform tasks. To perform different tasks in TDM, you require some Informatica privileges as well as different levels of Test Data Manager Service privileges.

You require the following PowerCenter Repository Service privileges to perform subset, masking, and generation tasks:

Privilege Group	Privileges
Runtime Objects	Create, Edit, and Delete Monitor Execute
Tools	Access Designer Access Repository Manager Access Workflow Manager Access Workflow Monitor
Design Objects	Create, Edit, and Delete
Sources and Targets	Create, Edit, and Delete
Folders	Create, Copy, and Manage Versions
Global Objects	Create Connections Manage Deployment Groups Execute Deployment groups Create Labels Create Queries

PowerCenter Repository Service Administrator Role to Generate and Run Workflows

To generate and run workflows, users must have the system-defined role **Administrator** on the PowerCenter Repository Service in addition to the required TDM privileges.

The following table describes each Test Data Manager privilege group:

Privilege Group	Description
Administration	Includes privileges to create and manage connections, pass phrases, roles and assign privileges to users and user groups from the Informatica Administrator, manage repositories, add licenses, and set up workflow and project attributes. **Note:** Before you can create users and groups, the default Informatica administrator user must assign Security Administration privileges to the Test Data Administrator user.
Data Domains	Includes privileges to view and manage data domains in the Test Data Manager.
Data Masking	Includes privileges to view and manage masking rules and policy assignments in the Test Data Manager.
Data Subset	Includes privileges to view and manage subset objects including entities, groups and templates in the Test Data Manager.
Policies	Includes privileges to view and manage policies in the Test Data Manager.
Projects	Includes privileges to view and manage projects, audit and import metadata, and execute plans and workflows in the Test Data Manager.
Rules	Includes privileges to view and manage masking and generation rules in the Test Data Manager.
Data Generation	Includes privileges to view and manage test data generation in the Test Data Manager.

For more information on Test Data Manager Service privilege groups, see the *Test Data Management Administrator Guide*.

Author

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